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$$z = \frac{1+i}{\sqrt{2}}$$

$$\mathbf{Teller} = 1+i$$

$$r_1 = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\tan \alpha_1 = \frac{1}{1} = 1 \Rightarrow \alpha_1 = \frac{\pi}{4}$$

$$\Rightarrow \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$\mathbf{Noemer} = \sqrt{2}$$

$$r_2 = \sqrt{0^2 + (\sqrt{2})^2} = \sqrt{2}$$

$$\tan \alpha_2 = 0 \Rightarrow \alpha_2 = 0$$

$$\Rightarrow \sqrt{2} \operatorname{cis} 0$$

$$\begin{aligned} z &= \frac{\sqrt{2} \operatorname{cis} \frac{\pi}{4}}{\sqrt{2} \operatorname{cis} 0} = \operatorname{cis} \left(\frac{\pi}{4} - 0 \right) \\ &= \operatorname{cis} \frac{\pi}{4} \end{aligned}$$

References